

CAMERA CULTURE GROUP
MIT MEDIA LAB

STARTUP GUIDE

SPAD Kit

MEMBERS WEEK · OCTOBER 2026

WELCOME



Thank you for spending Members Week with us. For almost two decades, Camera Culture has pushed the limits of what we can perceive by inventing new ways for cameras to see. Along the way we have filmed light in motion, looked around corners, and seen through fog and tissue. Single-photon sensors have been at the heart of much of it.

The kit in your hands is that same idea, small enough to plug into a laptop. We made it to give you a window into what we do. We hope you point it at problems we haven't thought of, and tell us what you find.

Ramesh Raskar

ASSOCIATE PROFESSOR
CAMERA CULTURE GROUP, MIT MEDIA LAB

THE TEAM

Meet the team



Nikhil Behari

Graduate student.



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Graduate student.



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Graduate student.



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SPAD WEBSITE

01 · OUR STORY

Measuring the time light takes to travel

2008 — Camera Culture founded
MIT MEDIA LAB

2009 — Looking around corners
ICCV · MARR PRIZE HM

2012 — Hidden 3D shape from a single wall
NATURE COMMUNICATIONS

2013 — Femto-photography
SIGGRAPH · TEST OF TIME

2015 — Tags located behind turbid layers
NATURE COMMUNICATIONS

2016 — Reading a closed book
NATURE COMMUNICATIONS

2018 — Photography through fog
ICCP

2023 — Two-bounce transients
CVPR

2024 — PlatoNeRF
CVPR

2025 — Navigating by hidden objects
ICRA

2026 — Hidden objects with consumer LiDAR
NATURE

2026 — Non-line-of-sight dataset using consumer SPADs
DENALI · CVPR

What we use it for

01

Non-line-of-sight

Seeing objects hidden around corners from light that bounces off a wall.

AROUND CORNERS · NATURE 2026 · DENALI

02

3D vision

Recovering full geometry, including mirrors and glass, from low-cost depth sensors.

PLATONERF · BLURRED LIDAR

03

Through scattering

Imaging through fog, tissue and layered materials by sorting photons by arrival time.

FOG · CLOSED BOOKS · SNOWPACK

04

Low-light

Perception from individual photon counts when there is almost no light.

05

High-speed

Capturing motion faster than conventional cameras can follow.

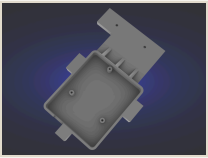
FEMTO-PHOTOGRAPHY

The kit



ST VL53L8CH

P-NUCLEO-53L8A1 expansion and development board



Printed mount

3D-printed housing for the ST sensor



USB cable

USB-A to Mini-B, connects the NUCLEO board to your laptop



Mini tripod

4–5 inch tabletop tripod to hold the sensor steady

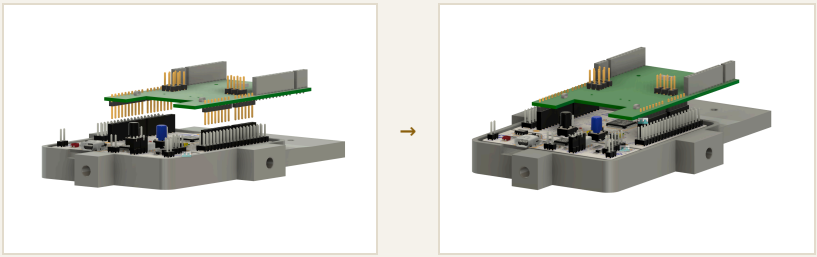


Your computer

Mac, Windows and Linux are all supported

Put it together

1 Attach Expansion Board



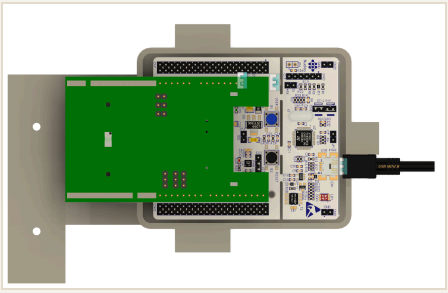
a. Align the header pins on the expansion board with the sockets on the dev board.

b. Plug the boards together.

2 Connect to Computer

a. Plug the USB Mini-B cable into the dev board.

b. Connect the other end to your computer.



04 · SOFTWARE SETUP

Set up your laptop

Open Terminal (PowerShell on Windows) and type each line, pressing Return after each one. You only do this once.

1 · DOWNLOAD THE SOFTWARE

```
git clone https://github.com/NikhilBehari/spad_capture.git
cd spad_capture
```

2 · RUN THE INSTALLER

MACOS · LINUX

```
./install.sh
```

WINDOWS

```
powershell -ExecutionPolicy Bypass -File install.ps1
```

3 · ACTIVATE THE ENVIRONMENT

On any system:

```
conda activate spad_capture
```

4 · FLASH YOUR BOARD

Connect the sensor to your laptop with the USB cable, then run the line for your board:

```
spad st  flash      # NUCLEO-F401RE + X-NUCLEO-53L8A1
spad tmf flash      # Arduino + TMF8828
```

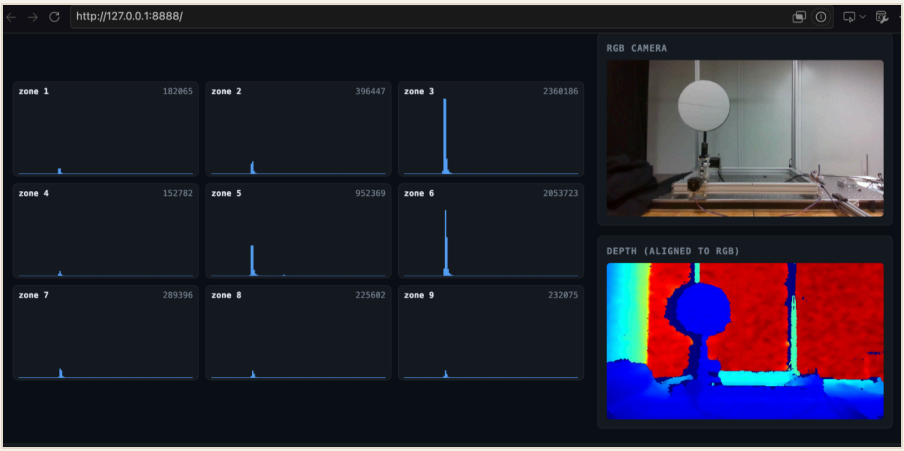
Visualize your captures

With the sensor plugged in, type:

```
spad st capture --viz
```

Then open this address in your web browser:

127.0.0.1:8888



Wave a hand in front of the sensor. The peak moves as the distance changes. Your capture is saved in the outputs folder.

What's next

Software and documentation

Every setting, zone selection and data format.

`github.com/NikhilBehari/spad_capture`

Datasets

DENALI and other captures from these sensors.

2027 challenge

Capture data with your kit and contribute to the dataset.

Scan the QR Code on the Back to See What Else You Can Do



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